MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 2

1. *Write* $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ *as a linear combination of* $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ *and* $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. *In how many different ways can* $\mathbf{u}$ *be written as a linear combination of* $\mathbf{v}$ *and* $\mathbf{w}$*?*

To write $\mathbf{u}$ as a linear combination of $\mathbf{v}$ and $\mathbf{w}$ we need to find the values of x_1 and x_2 such that the following equation holds:

$$x_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

The corresponding augmented matrix for this linear system of three equations and two variables is $\left[\begin{array}{cc|c} 1 & 1 & 3 \\ -1 & 1 & -1 \\ 0 & 1 & 1 \end{array}\right]$.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ -1 & 1 & -1 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{R2 \to R2+R1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{R3 \to R3-\frac{1}{2}R2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

From this echelon form we can deduce that the system is consistent (because there is no row of the form $[0 \dots 0|b]$ where $b \neq 0$) and has a unique solution (because there is no free variable). Therefore there is exactly one way to write $\mathbf{u}$ as linear combination of $\mathbf{v}$ and $\mathbf{w}$. Next we continue to get the row reduced echelon form.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R2 \to \frac{1}{2}R2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R1 \to R1-R2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

Thus, we get the unique solution $(x_1, x_2) = (2, 1)$, and therefore

$$2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

i.e. $\mathbf{u} = 2\mathbf{v} + \mathbf{w}$.

2. *Let* $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.

(a) *Is* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ *of the form* $A\mathbf{x}$ *for some* $\mathbf{x}$*? Set up the relevant system and solve it to verify your answer.*

Solutions will be posted at the end of Week 3.

We set up the system $Ax = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ *and row reduce.*

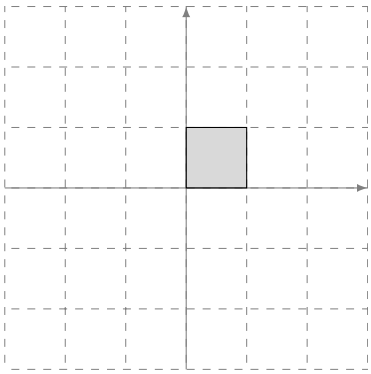
$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 2 & 2 & 1 \end{array}\right] \xrightarrow{R2 \to R2-2R1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$$

We see a row of $\left[\begin{array}{cc|c} 0 & 0 & 1 \end{array}\right]$ *so this system is inconsistent, and* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ *is not of the form* Ax *for any* x.

(b) *Now do the same as in* (a)*, but by thinking of vectors of the form* $A\mathbf{x}$ *as linear combinations of the columns of* A*. What form do such linear combinations take?*

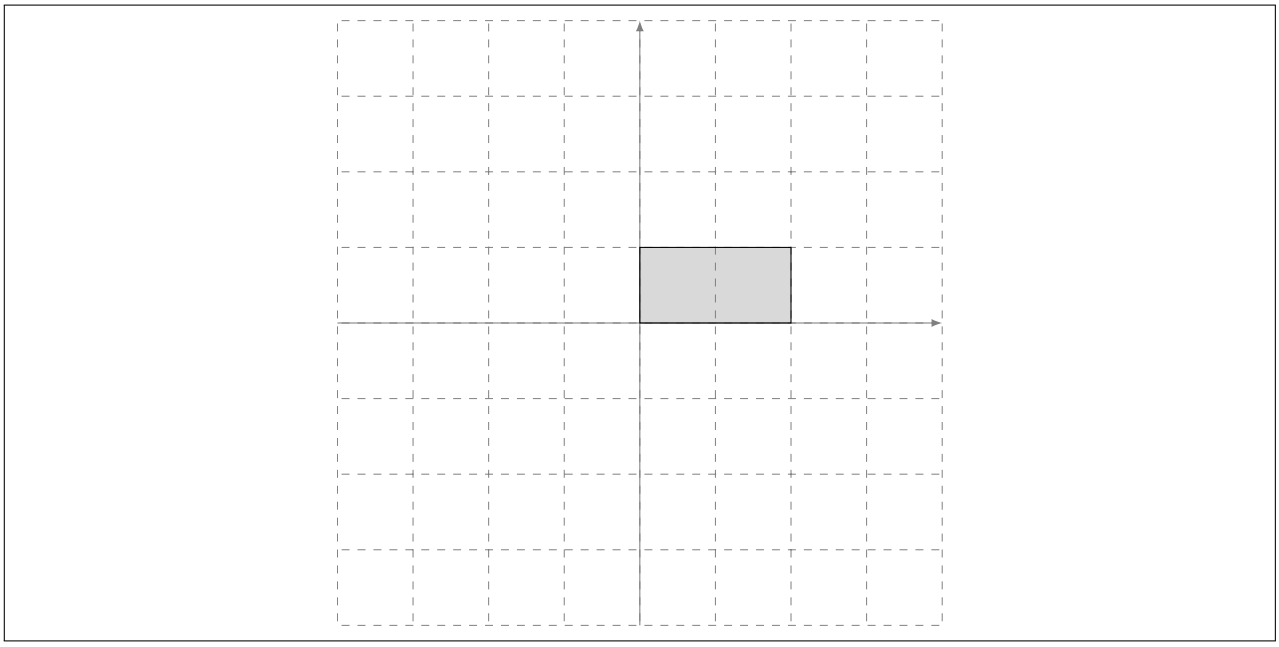
If $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$*, then* $Ax = x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = (x_1 + x_2) \begin{bmatrix} 1 \\ 2 \end{bmatrix}$*. Thus any vector of the form* Ax *is a multiple of* $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$*. Looking at the vector* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ *we see it is not of this form.*

3. *Consider the unit square in the plane:*

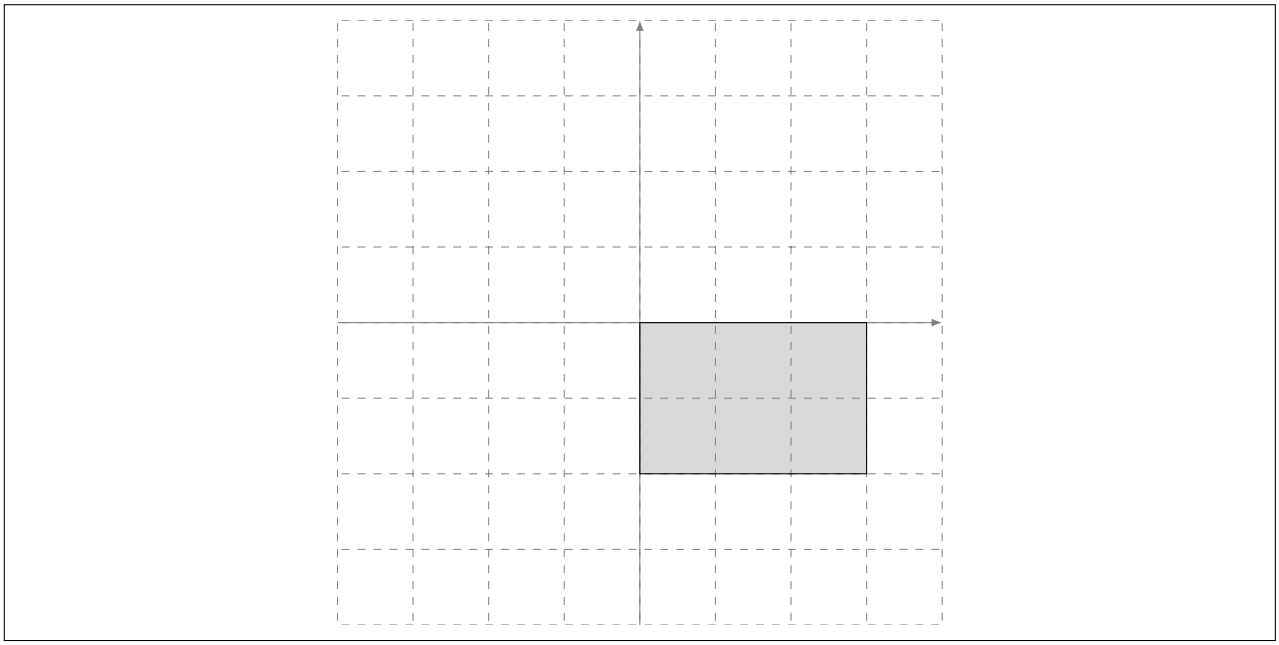


Suppose you take every vector $\mathbf{x} \in \mathbb{R}^2$ *corresponding to a point in the unit square and compute* $\mathbf{Ax}$ *for the given matrix* $\mathbf{A} \in \mathbb{R}^{2\times 2}$ *below. Which set of points do you obtain? (This means: Draw the set of all* $\mathbf{Ax}$*, where* $\mathbf{x}$ *is in the unit square.)*

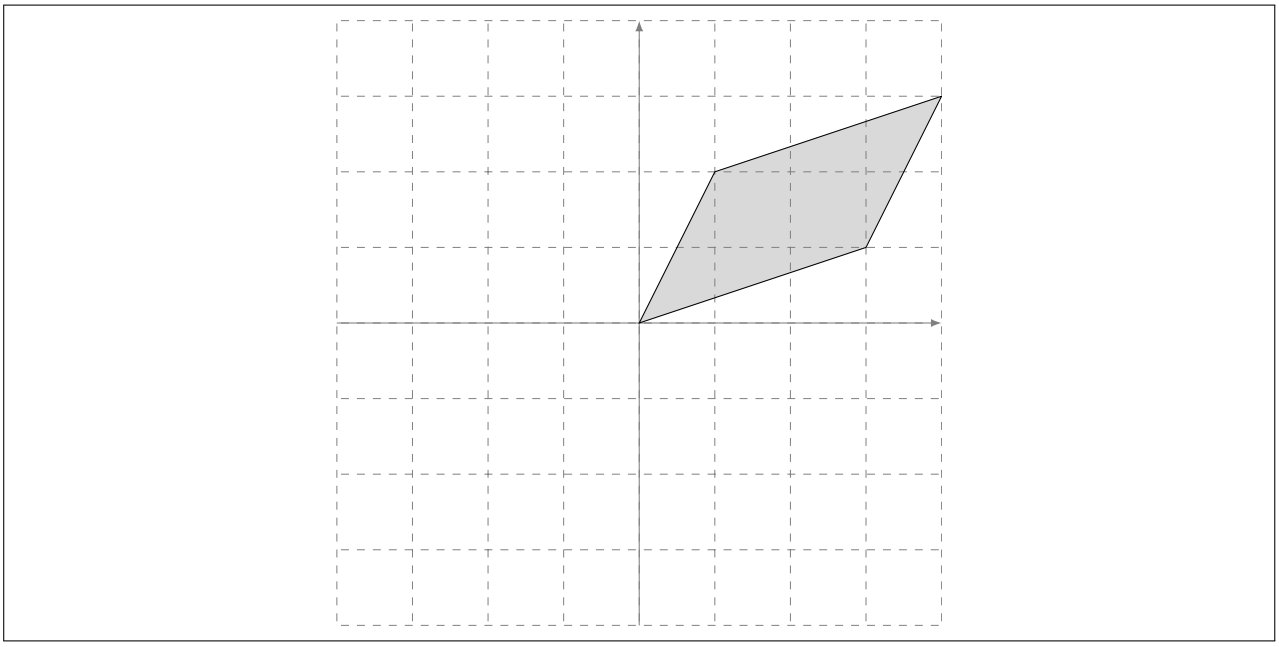
(a) $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$



(b) $A = \begin{bmatrix} 0 & 3 \\ -2 & 0 \end{bmatrix}$



(c) $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$



4. *Let θ be a real number. Then consider the following matrix:*

$$A_\theta = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

(a) *What does A_θ do to $\mathbb{R}^2$ geometrically? For a hint: consider what A_θ does to* $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

This matrix acts geometrically by rotating $\mathbb{R}^2$ by θ radians in the clockwise direction.

(b) *Using the geometric intuition from the previous part, show that $A_{\theta_1+\theta_2} = A_{\theta_1} A_{\theta_2}$*

If we rotate first by θ_2 radians clockwise, and then rotate by θ_1 radians clockwise, this is the same as rotating by $\theta_1 + \theta_2$ radians clockwise.

5. *Consider the following multiplication of two 3×3-matrices, where the question marks represent unknown coefficients:*

$$\begin{bmatrix} 6 & 6 & ? \\ 7 & ? & ? \\ 6 & 1 & 2 \end{bmatrix} \begin{bmatrix} ? & 1 & ? \\ 6 & 6 & 3 \\ ? & 5 & 5 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}.$$

With the information given, only one coefficient in the matrix on the right hand side can be calculated. Which one is it? Explain why and determine the value of this coefficient.

Only the third row of the first matrix and the second column of the second matrix are fully determined. Thus we can only compute c_{32}, which is $6 \cdot 1 + 1 \cdot 6 + 2 \cdot 5$.

6. *Let* $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. *Using what you know about elementary matrices, determine A^n for any natural number n.*

The elementary matrix A corresponds to the row operation that performs $R_1 \to R_1 + R_2$. Therefore A^2 corresponds to performing the row operation twice. In other words, we do the row operation $R_1 \to R_1 + 2R_2$. Moreover, for any $n \in \mathbb{N}$, A^n corresponds to the operation $R_1 \to R_1 + nR_2$. The elementary matrix for this operation is $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$.

7. *Consider the following four vectors in* $\mathbb{R}^3$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \mathbf{w}_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Is the span of $\mathbf{v}_1$ *and* $\mathbf{v}_2$ *equal to the span of* $\mathbf{w}_1$ *and* $\mathbf{w}_2$*? Explain why or why not!*

Observe that (one can also find such linear combinations by solving linear systems as what we did in Problem 1.)

$$\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}.$$

By the definition of the span, we know $\mathbf{w}_1, \mathbf{w}_2 \in \mathbf{Span}(\mathbf{v}_1, \mathbf{v}_2)$. Then all the linear combinations of $\mathbf{w}_1, \mathbf{w}_2$ are also in this span, that is, $\mathbf{Span}(\mathbf{w}_1, \mathbf{w}_2) \subseteq \mathbf{Span}(\mathbf{v}_1, \mathbf{v}_2)$.
Similarly, we have

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \qquad \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = -\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Then we know $\mathbf{v}_1, \mathbf{v}_2 \in \mathbf{Span}(\mathbf{w}_1, \mathbf{w}_2)$, and therefore $\mathbf{Span}(\mathbf{v}_1, \mathbf{v}_2) \subseteq \mathbf{Span}(\mathbf{w}_1, \mathbf{w}_2)$.
Finally, we can conclude that $\mathbf{Span}(\mathbf{v}_1, \mathbf{v}_2) = \mathbf{Span}(\mathbf{w}_1, \mathbf{w}_2)$.

8. *Consider the set*

$$W := \left\{ \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \in \mathbb{R}^3 \ : \ a_1 + a_2 - a_3 = 0 \right\}.$$

Can you write W *as a span of two vectors?*

First observe that W is the set of solutions for linear system $a_1 + a_2 - a_3 = 0$, with the corresponding augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \end{array}\right].$$

This is already in the row reduced echelon form. Note that a_2, a_3 are free variables. Therefore, we have

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} -a_2 + a_3 \\ a_2 \\ a_3 \end{bmatrix} = a_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Thus, W is the span of vectors $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

9. *Consider the following 2×2-matrices:*

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -3 \end{bmatrix},\ B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},\ C = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}.$$

Can you express A as a linear combination of B and C?

To write A as a linear combination of B and C we need to find the values of x_1 and x_2 such that the following equation holds:

$$x_1 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} + x_2 \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & -3 \end{bmatrix}.$$

This is equivalent to the following linear system of four equations and two variables,

$$\begin{aligned} x_1 + 3x_2 &= 1 \\ -x_1 - 2x_2 &= 0 \\ -x_1 + x_2 &= 3 \\ 2x_1 + x_2 &= -3 \end{aligned}$$

Write down the augmented matrix and do row reduce:

$$\left[\begin{array}{cc|c} 1 & 3 & 1 \\ -1 & -2 & 0 \\ -1 & 1 & 3 \\ 2 & 1 & -3 \end{array}\right] \xrightarrow{R4 \to R4+R3+R2} \left[\begin{array}{cc|c} 1 & 3 & 1 \\ -1 & -2 & 0 \\ -1 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R2 \to R2+R1,\ R3 \to R3+R1} \left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R3 \to R3-4R2} \left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R1 \to R1-3R2} \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

From here, we find a unique solution $(x_1, x_2) = (-2, 1)$, and this gives us that $A = -2B + C$.

10. *Suppose you are told there is a mystery 2×2-matrix A such that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.*

(a) Can you determine A just from this information?

Suppose that $A = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2\end{bmatrix}$, where both $\mathbf{a}_1$, $\mathbf{a}_2$ are vectors with two entries. Then we have

$$A\begin{bmatrix}1\\1\end{bmatrix} = \mathbf{a}_1 + \mathbf{a}_2 = \begin{bmatrix}-2\\2\end{bmatrix}, \qquad A\begin{bmatrix}-1\\1\end{bmatrix} = \mathbf{a}_1 - \mathbf{a}_2 = \begin{bmatrix}1\\-1\end{bmatrix}.$$

Sum these two equations together, we get

$$2\mathbf{a}_1 = (\mathbf{a}_1 + \mathbf{a}_2) + (\mathbf{a}_1 - \mathbf{a}_2) = \begin{bmatrix}-2\\2\end{bmatrix} + \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}-1\\1\end{bmatrix},$$

i.e. $\mathbf{a}_1 = \begin{bmatrix}-\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}$. Then we have $\mathbf{a}_2 = \begin{bmatrix}-2\\2\end{bmatrix} - \mathbf{a}_1 = \begin{bmatrix}-\frac{3}{2}\\ \frac{3}{2}\end{bmatrix}$. Therefore, we have $A = \begin{bmatrix}-\frac{1}{2} & -\frac{3}{2}\\ \frac{1}{2} & \frac{3}{2}\end{bmatrix}$.

(b) *Is always enough to know $A\mathbf{v}$ and $A\mathbf{w}$ for two distinct vectors $\mathbf{v}$ and $\mathbf{w}$ in order to determine A?*

No. Assume that $\mathbf{v} = \begin{bmatrix}1\\1\end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix}2\\2\end{bmatrix}$. Then similarly, we should obtain that

$$\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{b}, \qquad 2\mathbf{a}_1 + 2\mathbf{a}_2 = 2\mathbf{b},$$

for some constant vector $\mathbf{b}$. However, one of equations is obviously redundant so it is not enough to determine A.

11. *Let $A = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3\end{bmatrix}$ be a 3×3-matrix such that the columns of A sum up to the zero vector (i.e. $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$), and let $B = \begin{bmatrix}0 & 1 & 2\\ 0 & 1 & 2\\ 0 & 1 & 2\end{bmatrix}$. What can you say about the product matrix AB?*

Let $A = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3]$ such that $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}$. Then

$$AB = [0\mathbf{a}_1 + 0\mathbf{a}_2 + 0\mathbf{a}_3 \quad \mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 \quad 2\mathbf{a}_1 + 2\mathbf{a}_2 + 2\mathbf{a}_3]$$

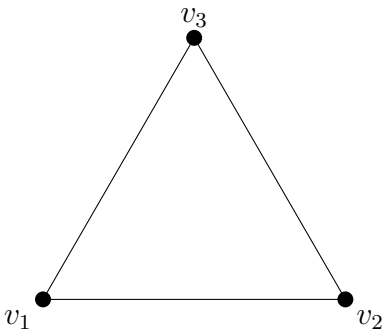
Since $2\mathbf{a}_1 + 2\mathbf{a}_2 + 2\mathbf{a}_3 = 2(\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3) = \mathbf{0}$, the product AB has to be the zero matrix.

1. Applications

12. *In network theory and data science (e.g. while studying how information flows in social networks), graphs are used to represent a given network. A graph is a set of vertices (nodes) and edges (links) connecting them. There is a matrix, called the* **adjacency matrix**, *which encodes the information of how vertices are connected. It is defined as follows: if the network has V vertices, then the adjacency matrix A is a $V \times V$-matrix such that*

$$A(i,j) = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are connected by an edge} \\ 0 & \text{otherwise.}\end{cases}$$

Here $A(i,j)$ denotes the entry of the matrix in the i-th row and j-th column. For example, if the network has two nodes and one edge connecting them, the adjacency matrix is $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$. Consider the case where the network is given by a triangle (see Figure), where there are three vertices and vertices connecting each pair of them.



(a) *Compute the adjacency matrix A. What can we say about A and A^T?*

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = A^T.$$

The fact that the adjacency matrix and its transpose are the same is a general fact. This follows from the fact that, if v_i and v_j are connected by an edge, then clearly v_j and v_i are also connected. Therefore $A = A^T$.

(b) *Compute A^2 and A^3. What is the meaning of the entries of these two matrices? (Hint: The numbers in the entries are counting the number of ways of doing a certain task which involves vertices and edges.)*

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, A^3 = \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix}.$$

Looking at the values of A^2 and A^3 it is possible to note the following pattern. In the diagonal we get the same values and such numbers are counting the number of paths which start and end at a given vertex, by suing two edges (in A^2) and three edges (in A^3). This also occurs for the other entries. In general $A^k(i,j)$ is the number of paths between v_i and v_j using k-edges.

(c) *Suppose you are the operator of a large network of railroad stations whose adjacency matrix is A (two stations are connected by an edge if there is a direct connection between the two stations). Can you come up with an algorithm that uses just the matrix A to determine whether you can reach every station from every other station in your network? Is there way to determine the minimal k such that you can get from any station to any another station with passing fewer than k stations along the way?*

For an integer k, define the matrix $S_k = A + A^2 + \cdots + A^k$. By (b), the entry $S_k(i,j)$ represents the number of paths between v_i and v_j using at most k edges. Suppose that at some stage k, all entries of S_k are nonzero. Then this implies that for every pair of vertices, there exists at least one path between them (with at most k edges), i.e. they are reachable from each other.

Therefore, we start with $S_1 = A$ and iteratively compute S_k. Once we find a matrix S_k such that all entries are nonzero, then we stop the algorithm and conclude that can reach every station from every other station, and the minimal k is indeed the smallest integer such that all entries of S_k are nonzero. If we could not find such S_k after n steps, then it implies that we cannot reach every station from some of the stations.